# Software Engineering

# Project Proposal:

## FAST Sports Complex Management System

**Spring 2026**

**SECTION BCS-6B**

**SUBMITTED BY**

**Gotam Dulhani – 23I-0545**

**Abdul Saboor – 23I-3039**

**SUBMITTED TO**

**Engr. Abdul Rahman**

**SUBMISSION DATE**

**16th Feb 2026**

**DEPARTMENT OF COMPUTER SCIENCE,**

**FAST -NUCES, KARACHI**

**FAST Sports Complex Management System (FSCMS)**

# 1. Overview

The FAST Sports Complex currently manages facility bookings, tournaments, equipment, and payments manually or through fragmented systems. This leads to scheduling conflicts, lack of transparency, poor resource tracking, and inefficient communication between students, faculty, coaches, and administration.

The customer for this project is **FAST NUCES University Sports Department**. The system will serve students, faculty, coaches, sports coordinators, and administrative staff.

This project will deliver a **new centralized web-based Sports Complex Management System**, not an extension of the existing system. It will digitize bookings, tournaments, equipment, payments, and analytics into one integrated platform.

Estimated development duration: **2-3 months**.

This project may later integrate with FAST's student portal or ERP system.

# 2. Project Detail

## 2.1 Problem or Challenge

At FAST, sports facilities such as badminton courts, football grounds, and basketball courts are heavily used. However, bookings are often handled manually or informally, causing double bookings, unfair slot distribution, and disputes. There is no centralized system to manage tournaments, track equipment inventory, monitor peak hours, or maintain facility history.

Students often face uncertainty regarding facility availability. Equipment loss and damage go untracked. Tournament organization requires manual coordination. Revenue tracking and reporting are inefficient.

This lack of automation reduces transparency, operational efficiency, and overall sports engagement within the university.

## 2.2 Project Vision

Our vision is to develop a **smart, centralized, and automated Sports Complex Management System** that ensures fair access, efficient scheduling, transparency, and data-driven decision-making.

The system will streamline bookings, tournaments, equipment tracking, payments, analytics, and communication — creating a modern digital sports ecosystem at FAST.

Success will be measured by reduced booking conflicts, improved facility utilization, automated reporting, and enhanced user satisfaction.

## 2.3 Project Feasibility Study

### 2.3.1 Technical Feasibility

- Web-based system using modern technologies (React, Node.js / Django, MySQL/PostgreSQL).
- Cloud or university-hosted deployment.
- Developers possess required programming and database skills.
- Integration capability with university authentication systems.
- No specialized hardware required.

Technical feasibility: **High**

### 2.3.2 Operational Feasibility

- Sports department staff will operate admin dashboard.
- Students and faculty already use digital portals, so adoption barrier is low.
- System reduces manual workload.

Operational feasibility: **High**

### 2.3.3 Economic Feasibility

- Development cost: minimal (student project).
- Hosting: university server or low-cost cloud.
- Long-term savings from automation and reduced equipment loss.
- Improved revenue tracking from facility booking.

Economic feasibility: **Cost-effective and beneficial**

## 2.4 Project Goals

**Functional Goals**

- Online facility booking
- Tournament management
- Equipment inventory tracking
- Payment processing
- Coach scheduling
- Analytics dashboard

**Technological Goals**

- Secure role-based authentication
- Scalable modular architecture
- Real-time availability updates
- Notification system integration

**Quality Goals**

- Reliable booking conflict detection
- High system availability
- Accurate reporting and analytic

**Organizational Goals**

- Digital transformation of sports department
- Reduce manual paperwork
- Improve transparency

**Other Goals**

- High usability
- Mobile-responsive design
- Data portability

**Constraints**

- University IT policies
- Limited project duration (one semester)
- Budget limitations

- Data privacy regulations

## 2.5 Project Abstract

The FAST Sports Complex Management System (FSCMS) is a comprehensive web-based solution designed to digitize and automate the management of sports facilities at FAST NUCES. Currently, sports facility bookings, tournament organization, equipment tracking, and revenue management are handled manually or through fragmented processes, leading to inefficiencies, scheduling conflicts, and lack of transparency.

FSCMS aims to provide an integrated platform that enables students and faculty to reserve facilities such as badminton courts, football grounds, basketball courts, table tennis tables, and pool tables through an automated time-slot-based booking system. The system will include real-time availability checking and conflict detection to prevent double bookings.

The system will also support tournament management including team registration, match scheduling, bracket generation, score tracking, and leaderboards. Equipment management features will track inventory, check-outs, maintenance schedules, and low-stock alerts.

Additional modules include payment and billing management, walk-in queue handling, coach scheduling, inter-department competition tracking, event management, notification systems, analytics dashboards, and maintenance issue reporting.

Administrators will have access to dashboards displaying facility utilization patterns, peak hours, revenue reports, and user activity logs. Role-based access control will ensure appropriate permissions for students, staff, coaches, and administrators.

The project will be developed using a layered architecture approach and Agile methodology to ensure incremental delivery and continuous improvement. The final outcome will be a scalable, secure, and user-friendly system that modernizes sports management at FAST while promoting fairness, efficiency, and engagement.

## 2.6 Project Scope

**Included:**

- Facility booking system
- Tournament management
- Equipment tracking
- Payment & billing
- Coach management

- Analytics dashboard
- Maintenance reporting
- Notification system
- Inter-department competition tracking

**Not Included:**

- Physical access control hardware
- Biometric entry systems
- Integration with external banking APIs (mock payments only for academic project)
- Mobile native app (web-based only)

## 2.7 Project Team

| Organization | Name | Availability | Comment |
|---|---|---|---|
| FAST | Abdul Saboor | Full-time | Backend & Database |
| FAST | Gotam Dulhani | Full-time | Frontend & Testing |

## 2.8 Existing & Related Work

| System | Features | Limitations | Our Uniqueness |
|---|---|---|---|
| University ERP Sports Module | Basic booking | No tournament & equipment integration | Complete ecosystem |
| Generic Booking Apps | Slot booking | Not sports-specific | Domain-focused |
| Google Calendar | Scheduling | No conflict logic for facilities | Automated sports logic |

Our uniqueness:

- Integrated tournament + booking + equipment
- Inter-department competition tracking

- Sports analytics & revenue insights
- Queue management with priority rules

## 2.9 Proposed Solution & Architecture

## Architecture: 3-Layer Architecture

**1. Presentation Layer**

- Web UI (Student, Coach, Admin dashboards)

**2. Application Layer**

- Booking Engine
- Tournament Engine
- Equipment Manager
- Payment Module
- Analytics Module
- Notification Service

**3. Data Layer**

- Centralized relational database
- Logs and transaction storage

**2.10 Use Case Model**

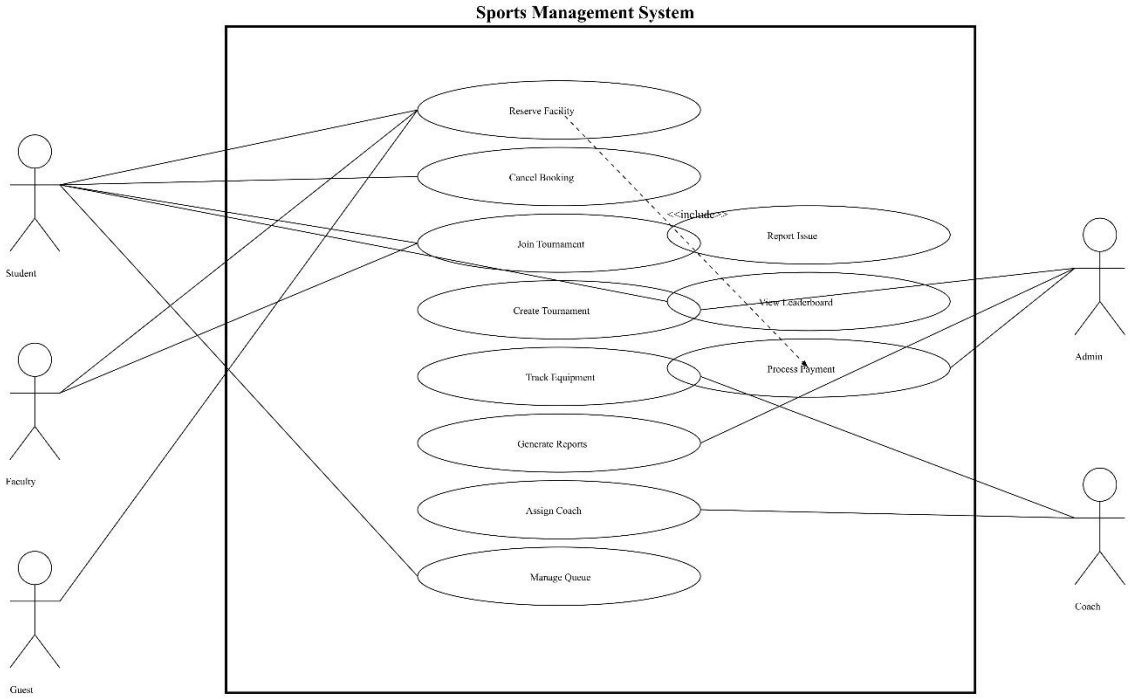
**Actors:**

- Student
- Faculty
- Admin
- Coach
- Guest

**Major Use Cases:**

- Reserve facility
- Cancel booking
- Join tournament

- Create tournament (Admin)
- Track equipment
- Generate reports
- Assign coach
- Manage queue
- Report issue
- View leaderboard
- Process payment




## 2.11 Feature List (20 Core Domain Features)

1. Facility reservation with time-slot management
2. Real-time availability checking

3. Booking conflict detection
4. Tournament creation
5. Match scheduling & bracket generation
6. Score tracking & leaderboard generation
7. Equipment inventory tracking
8. Equipment check-in/check-out system
9. Maintenance scheduling
10. Walk-in queue management
11. Priority booking rules
12. Membership package management
13. Fine management system
14. Coach session scheduling
15. Inter-department points tracking
16. Event management module
17. Facility usage analytics
18. Revenue analytics
19. Issue reporting & tracking
20. Cancellation & refund processing

## 2.12 Software Development Methodology

## Agile Scrum Methodology

Reason:

- Large feature set
- Modular system
- Continuous feedback from sports department

Plan:

- Sprint duration: 2 weeks
- Sprint 1: Booking Module

- Sprint 2: Tournament Module
- Sprint 3: Equipment & Payments
- Sprint 4: Analytics & Notifications
- Sprint 5: Testing & Deployment

## 2.13 Proposed Infrastructure Platform & Tools

- React.js
- Node.js / Express OR Django
- MySQL / PostgreSQL
- REST API
- JWT Authentication
- Git & GitHub
- Postman
- Draw.io
- Papyrus UML
- Balsamiq
- MS Project
- VS Code
- Docker (optional)
- Linux Server / University Server

***Note: Some technologies could change, according to the requirements and convenient to the application***

## 2.14 Project Timeline

Month 1:

- Requirement analysis
- System design
- Use case modeling

Month 2:

- Booking & user management implementation

Month 3:

- Tournament & equipment modules

Month 4:

- Payments, analytics, notifications and Testing
- Documentation
- Deployment

## 2.15 Mockups

Suggested Screens:

1. Facility Booking Calendar View
2. Admin Dashboard (Analytics + Revenue)
3. Tournament Bracket View
4. Equipment Inventory Screen
5. User Profile & Booking History

## 2.16 References

[1] I. Sommerville, *Software Engineering*, 10th ed. Pearson, 2016.
[2] R. Pressman, *Software Engineering: A Practitioner's Approach*, 8th ed. McGraw-Hill, 2014.
[3] FAST NUCES Sports Policy Document, 2025.